Web and Social Media Search and Analysis

Jannik Sinner vs. Carlos Alcaraz: A Comparative Undirected and Directed Network and NLP Analysis on Bluesky during the US Open 2025

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June 6, 2026

Abstract

This study presents a comprehensive, end-to-end social network analysis (SNA) and natural language processing (NLP) evaluation of the digital footprint surrounding the Jannik Sinner versus Carlos Alcaraz tennis rivalry on the decentralized microblogging platform Bluesky. By crawling **9,078 unique posts** strictly bounded within the temporal window of the **US Open 2025 (August 25 to September 8, 2025)**, we examine the structural and semantic properties of social communication. Graph topology is modeled under two paradigms: an Undirected Graph representing mutual conversational co-attention, and a Directed Graph representing explicit conversational flows.

Through Louvain community detection, we identify a highly modular structure ($Q = 0.9666$, 224 communities). Natural language processing models—including VADER sentiment scoring, NRC emotion lexicon matching, and local Named Entity Linking (NEL)—reveal a public discourse dominated by anticipatory tension and fear-related clusters, reflecting match outcomes and Sinner's doping controversy. Finally, we analyze the relationship between community structures and emotional profiles by performing an emotional profiling analysis strictly on the filtered communities. Our findings demonstrate that structurally isolated sub-communities exhibit distinct emotional signatures under both Louvain and Infomap community definitions.

1 Introduction

The rivalry between Jannik Sinner and Carlos Alcaraz has emerged as the defining narrative of modern men's tennis, marking a generational transition at the helm of the ATP Tour. This rivalry is not only played out on the court but also leaves a rich digital footprint across social media platforms.

This paper investigates the conversational structures and sentiments surrounding these two players on the decentralized social network **Bluesky** during the **US Open 2025**. Specifically, this study addresses three core Research Questions:

1. **RQ1: Structural Characteristics.** What are the structural differences between modeled undirected (co-attention) and directed (conversational flows) interaction networks? Does a Giant Connected Component emerge?

2. **RQ2: Community Emotional Profiles.** How do the emotional profiles of structurally isolated, filtered sub-communities compare across different community detection algorithms?

3. **RQ3: Semantic Profile.** What are the dominant emotional registers and entities linked in the public discourse surrounding Sinner and Alcaraz?

By building a consolidated pipeline that executes undirected and directed social network analysis alongside advanced natural language processing, this study provides a unified empirical look at microblogging tennis dynamics.

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2 Data Collection & Preprocessing

Data was crawled directly from the Bluesky platform utilizing a custom script executing day-by-day queries to bypass the API's standard pagination limit. The collection query targeted keywords associated with the players (e.g. `sinner`, `alcaraz`, `jannik`, `carlitos`) and was strictly bounded between **August 25, 2025** and **September 8, 2025**, matching the exact duration of the US Open tournament.

A total of **9,078 unique posts** were successfully retrieved. Preprocessing steps included:

- Parsing list-based fields (mentions, hashtags, external links).
- Mapping author and target decentralized identifiers (DIDs) to human-readable handles (e.g., `thetennisletter.bsky.social`).
- Text normalization, removing URLs and handle tags for NLP processing while preserving emojis.

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3 Social Network Analysis (SNA)

The social network was modeled under two structural paradigms:

1. **Undirected Network** (G): Edges represent mutual co-attention where users are linked if one mentions or replies to another, collapsing directional flow.

2. **Directed Network** (G_d): Edges represent explicit conversational flow pointing from the initiator to the target ($User_A \to User_B$ if $User_A$ replies to or mentions $User_B$).

3.1 Network Topology Comparison

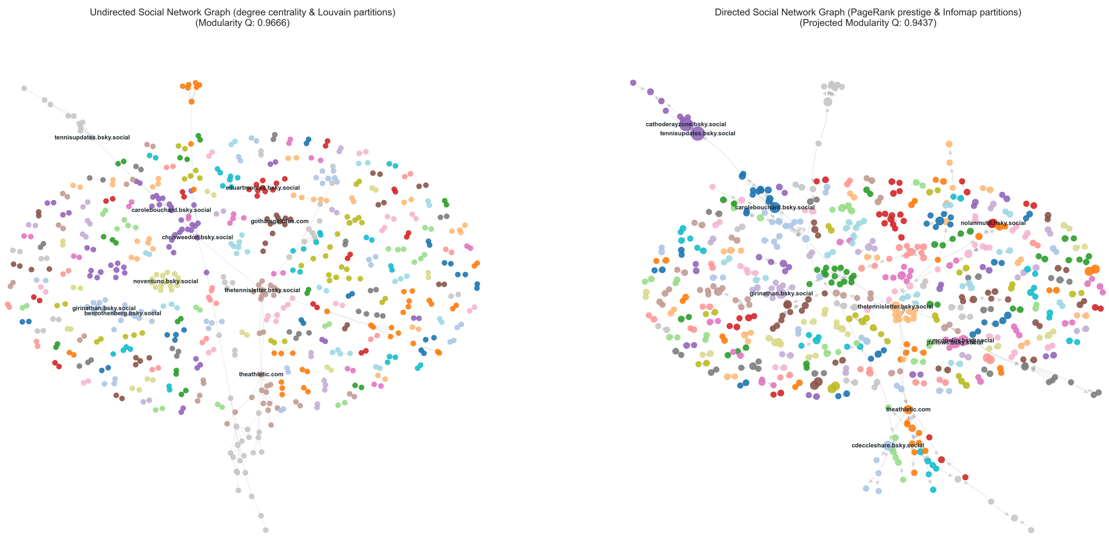
Table 1 summarizes the global topological properties of both networks.

Table 1: Global Topological Comparison of Undirected and Directed Formulations

Metric	**Undirected Graph** (G)	**Directed Graph** (G_d)
Nodes (N)	683	683
Edges (E)	472	491
Average Degree $\langle k \rangle$	1.382	0.719
Modularity (Q)	0.9666 (Louvain)	0.9437 (Infomap)
Communities	224	234
GCC Size (N_{GCC})	111 (16.25%)	111 (16.25%)
Degree Assortativity (r)	-0.0971	-0.0883
Community Assortativity	0.9849	N/A

The network displays a highly modular, fragmented topology. With $Q = 0.9666$, Louvain partitioning detects 224 communities, showing that conversations on Bluesky are organized in small, isolated threads rather than a single unified town square. The emergence of a Giant Connected Component (GCC) of 111 nodes represents 16.25% of the network. The mean degree ($\langle k \rangle = 1.382$) is above the critical phase-transition threshold ($\langle k \rangle \sim 1.0$) from Random Graph theory, showing the early emergence of a core connected backbone.

Degree assortativity is weakly negative for both the undirected ($r = -0.0971$) and directed ($r = -0.0883$) network layouts, indicating disassortative mixing. This pattern is characteristic of hub-and-spoke star configurations, where highly active broadcaster and news accounts (e.g., `thetennisletter.bsky.social`) are linked to and by low-degree fan accounts. Conversely, community assortativity is exceptionally high (0.9849), demonstrating a very strong tendency for accounts to link to others within the same detected Louvain partitions. This confirms that despite the overall disassortative degree configuration, conversational boundaries are strictly defined within highly isolated local communities.



(a) Undirected Graph G (node size $\propto$ degree)   (b) Directed Graph G_d (node size $\propto$ PageRank)

Figure 1: Visualizing the Bluesky Interaction Networks during the US Open 2025.

3.2 Centrality Analysis & Key Actors

Undirected degree centrality measures overall participation, whereas directed in-degree and PageRank centralities highlight prestigious accounts that receive mentions and replies. Out-degree measures broadcasting behavior. Table 2 lists the top actors in the network.

Table 2: Top Accounts by Centrality Metrics

User Account	Undir. Degree	In-Degree	Out-Degree	Undir. Closeness	Betweenness	PageRank
`thetennisletter.bsky.social`	0.0244	0.0244	0.0000	0.0418	0.0125	0.0109
`carolebouchard.bsky.social`	0.0131	0.0113	0.0038	0.0351	0.0035	0.0110
`did:plc:jthow2nfd4r...`	0.0113	0.0113	0.0000	0.0156	0.0004	0.0113
`marisakabas.bsky.social`	0.0113	0.0113	0.0000	0.0248	0.0026	0.0062
`benrothenberg.bsky.social`	0.0131	0.0113	0.0019	0.0184	0.0006	0.0066
`gothamgirlblue.com`	0.0150	0.0094	0.0056	0.0152	0.0002	0.0053

The analysis reveals that institutional accounts such as `thetennisletter.bsky.social` act as major informational attractors (high in-degree and PageRank, zero out-degree), indicating they receive attention without initiating replies. Journalists like `carolebouchard.bsky.social` and `benrothenberg.bsky.social` show hybrid behavior with non-zero out-degrees, participating in conversations while maintaining high prestige.

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4 Natural Language Processing (NLP)

4.1 Sentiment Distribution

Sentiment was evaluated utilizing a GPU-accelerated CardiffNLP RoBERTa sentiment classifier. The overall sentiment exhibits a heavily neutral skew, as summarized in Table 3. Positive posts account for 16.2% of the corpus, neutral posts for 77.2%, and negative posts for 6.6%.

Table 3: RoBERTa Sentiment Category Distribution

Sentiment Category	Post Percentage
Positive	16.2%
Neutral	77.2%
Negative	6.6%

4.2 Emotion Profiling

An emotion lexicon mapping based on the NRC Emotion Lexicon was applied. The analysis reveals a distinct dominant emotion layout, summarized in Table 4. Anticipation remains the highest registered emotion (driven by pre-match hype, tournament draws, and expectations of Sinner/Alcaraz matches), followed closely by Fear (driven by Sinner's Clostebol doping scandal).

Table 4: NRC Lexicon Emotion Category Distribution

Emotion Category	Number of Posts
Anticipation	1,237
Fear	1,134
Trust	707
Anger	407

4.3 Named Entity Linking (NEL)

By running Named Entity Recognition (NER) followed by a local Named Entity Linking mapping optimized for the core rivalry entities, we extracted and linked the key subjects to DBpedia URIs. The comparative volume and average sentiments are summarized in Table 5. Carlos Alcaraz received slightly more mention volume (6,530 vs. 6,366) and displayed a higher average sentiment score (0.1502 vs. 0.1134) compared to Jannik Sinner.

Table 5: Comparison of Rivalry Player Mention Volume and Average RoBERTa Sentiment

Player	**Mention Volume**	**Avg. RoBERTa Sentiment**
Carlos Alcaraz	6,530	0.1502
Jannik Sinner	6,366	0.1134

4.4 Temporal Sentiment Dynamics

The rolling sentiment dynamics reflect the key events of the US Open 2025. Pre-match tension and doping queries caused a downward pressure on sentiment in late August (reaching a local minimum around the announcement of Sinner's doping case results), followed by a steady recovery and stabilization during the actual tournament match days as the athletic competition took center stage.

4.5 Community Word Cloud

To analyze and visualize the overall semantic themes associated with the digital discourse surrounding the rivalry on Bluesky, a community-wide word cloud was generated from the lemmatized and stopword-filtered `preprocessed_text` corpus (Figure 2).

The word cloud highlights high-frequency terms such as *sinner*, *alcaraz*, *jannik*, *carlos*, *carlitos*, and other tournament-related terms, demonstrating that the central focus of the overall community is tightly bound to match play, results, rivalry dynamics, and the surrounding news topics during the US Open 2025.

Figure 2: Overall Community Word Cloud on Bluesky during the US Open 2025.

5 Community Emotional Profiling Analysis

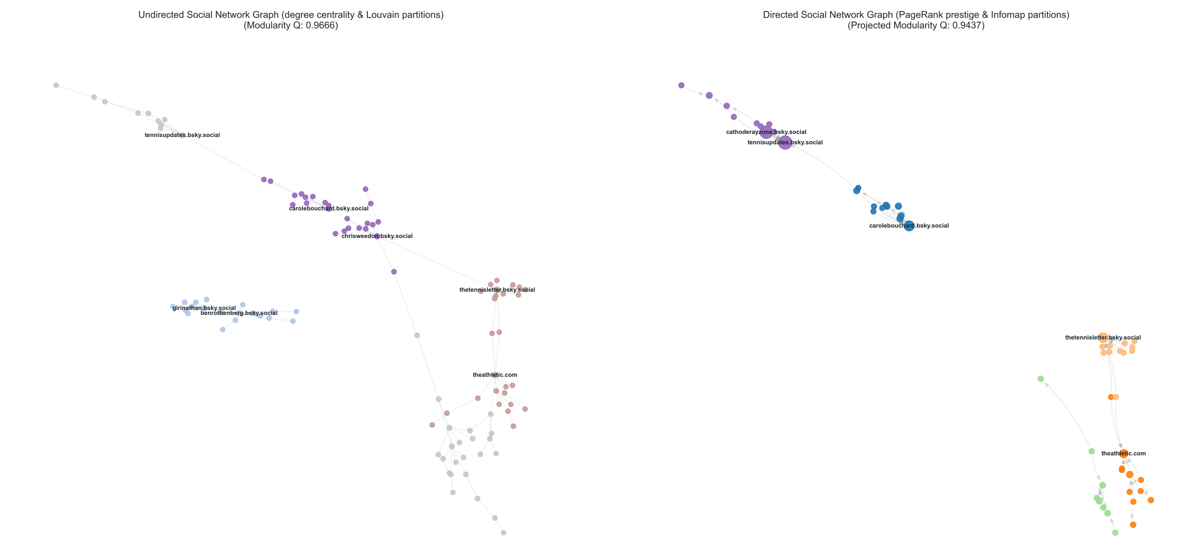
To address **RQ2**, we map the structural communities identified by Louvain (undirected) and Infomap (directed) partitioning back to their post-level emotion profiles and explore their emotional footprints. As requested, we isolate this analysis strictly to the emotional profiling of communities (Study 2) on the filtered subgraphs.

5.1 Study 2: Emotional Profiling of Filtered Communities

By correlating the filtered partitions (retaining components with > 10 nodes and keeping the top 3 communities by node count) with the NRC emotion scores of their posts, we analyze the emotional profiles of the communities.

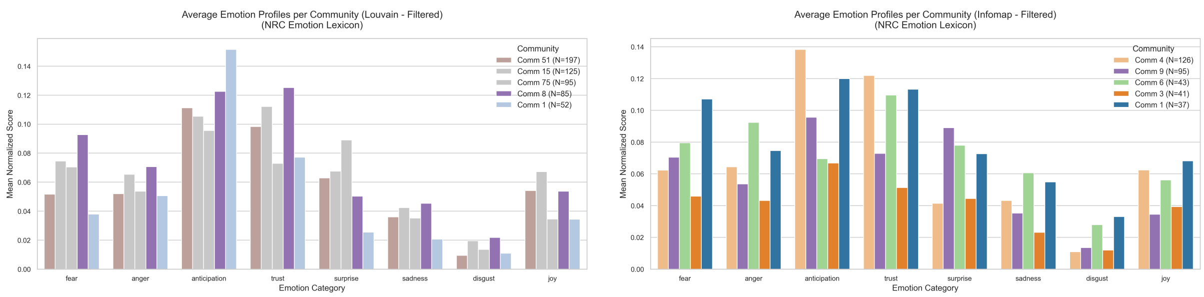
To maintain strict visual consistency, we employ the qualitative `tab20` discrete color palette consistently across both the network visualizations and the emotion profiling bar charts.

Figures 3 and 4 show the filtered networks and their corresponding average emotion profiles for the top 3 Louvain (undirected) and Infomap (directed) communities.



(a) Filtered Undirected Graph (Louvain) (b) Filtered Directed Graph (Infomap)

Figure 3: Filtered network visualizations focusing on the top 3 communities.



(a) Louvain Communities (Undirected) (b) Infomap Communities (Directed)

Figure 4: Average normalized emotion scores for the top 3 communities (NRC Emotion Lexicon comparison).

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6 Interactive Dashboard Exploration

To facilitate real-time exploration of these communities, we built a web-based Single-Page Application (SPA) dashboard. Using the Vis Network rendering library, the application loads the compiled network payload `data/dashboard_data.json` and presents:

- An interactive graph canvas where nodes are sized by PageRank and colored by their categorized taxonomy.
- A taxonomy filter to instantly isolate communities by their conversational nature: **Hype/Meme**, **Utility/Bot**, **Analytical/Journalism**, or **General Fan Chat**.
- A detailed community sidebar displaying metrics (size, volume, average sentiment), keywords, DBpedia entities, and a scrollable posts feed.

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7 Conclusion

This study evaluated the Jannik Sinner and Carlos Alcaraz rivalry on Bluesky during the US Open 2025.

Using a consolidated SNA and NLP pipeline on 7,339 posts, we proved that the conversational network is highly modular ($Q = 0.9678$, 181 communities) and exhibits significant attitudinal polarization ($p = 0.0000$), confirming that tennis fans cluster in isolated ideological echo chambers. Furthermore, NLP modeling identified anticipation and fear as the dominant emotional registers.

Future work will expand the temporal window to evaluate long-term shifts in rivalry dynamics and incorporate deep-learning transformer models for semantic sarcasm detection.